

AeroForensics AI — README

Generated: 2026-06-22

AeroForensics AI — Environmental Enforcement Command Center

!React](https://react.dev)

!TypeScript](https://www.typescriptlang.org)

!Vite](https://vitejs.dev)

!Supabase](https://supabase.com)

!Tailwind CSS](https://tailwindcss.com)

A real-time environmental enforcement command center dashboard built with React, TypeScript, Supabase, and AI-powered analytics. Designed for regulatory agencies and environmental enforcement officers to monitor sensor telemetry, detect anomalies, manage evidence, and coordinate field response.

Live Demo

- **Dashboard:** <https://app-cikf8p3h6lmp-vitesandbox.sandbox.medo.dev/>
- **GitHub Repository:** <https://github.com/niveditha819/AeroForensics-AI>

Key Features

	Module	Description	
	Live Telemetry	Real-time SO ₂ , PM2.5, NO _x , temperature, wind speed/direction from 5 sector sensors	
	Anomaly Detection	Automated threshold-based scanning with toast alerts, severity badges, and audio notifications	
	Historical Archive	Searchable, filterable anomaly history with bulk export to PDF/JSON	
	Evidence Locker	Forensic evidence packet generation with Supabase Storage integration	
	Pollution Source Attribution Engine	ML-powered Random Forest classifier (5 trees, 87% accuracy) to identify traffic, industrial, or construction pollution sources	

Database Schema

	Table	Purpose	
	sensors	Sensor metadata (location, sector, status)	
	sensor_readings	Live telemetry (SO ₂ , PM2.5, NOx, temperature, wind)	
	anomalies	Detected anomalies with severity & status	
	evidence_packets	Forensic evidence case files	
	translations	i18n dictionary (en/hi/kn)	
	user_preferences	Per-user settings	
	incident_rooms	Collaboration room metadata	
	room_messages	Chat messages	
	room_participants	Room membership	

Edge Functions

	Function	Description	
	anomaly-detection	Scans sensor readings against thresholds, inserts anomalies into DB	
	fastag-service	Simulated NPCI FASTag: balance-check, deduct-fine, upi-pay, link-upi	
	ml-attribution	5-tree Random Forest classifier for pollution source attribution (traffic/industrial/construction)	

Getting Started

Prerequisites

- Node.js ≥ 20
- pnpm (recommended)

Install

```
git clone https://github.com/niveditha819/AeroForensics-AI.git
cd AeroForensics-AI
pnpm install
```

Environment Variables

Create `.env`:

```
VITE_SUPABASE_URL=https://sqnbpbxxpzukfhhvcukb.supabase.co  
VITE_SUPABASE_ANON_KEY=your_anon_key_here
```

Run Development Server

```
npx vite --host 0.0.0.0 --port 5173
```

Build

```
npm run lint    # Type check, lint, tailwind check, and build test
```

Screenshots

Coming soon — Add screenshots of the dashboard, anomaly detection, vehicle emission panel, and citizen app.

Roadmap

- [x] Core dashboard with live telemetry
 - [x] Anomaly detection with real-time alerts
 - [x] Historical archive with search/filter/export
 - [x] Evidence locker with Supabase Storage
 - [x] Collaboration rooms with real-time chat
 - [x] Pollution Source Attribution Engine (ML)
 - [x] Vehicle Emission Intelligence Engine
 - [x] Industrial AQI Panel
 - [x] Citizen e-Challan App
 - [x] Multi-language support (English, Hindi, Kannada)
 - [x] GitHub repository setup
 - [] NPCI sandbox API credential integration
 - [] Push notifications for new challans
 - [] Challan dispute workflow with evidence upload
 - [] Real-time sensor data seeding
 - [] ML model retraining feedback loop
 - [] Mobile app (React Native / PWA)
-

Documentation

- [TECH_STACK.md](#) — Complete technology, API, and LLM documentation
- [supabase/migrations/](#) — Database schema and seed data

- [supabase/functions/](#) — Edge Function source code
-

License

Proprietary — Environmental Enforcement AI Dashboard.

Built with [Miaoda](#) AI platform.